CHE 313: Transport Phenomena III

Instructor: Sangil Kim (312-355-5149, sikim@uic.edu), Chemical Engineering
Spring 2020: Tuesday and Thursday, Thomas Beckham Hall Rm# 180G, 11:00 am -12:15 pm

Course Aims

Introduction to the fundamentals of separation processes and their applications in a variety of technologies (i.e., distillations, absorption, extraction, membranes). Includes methods for the analysis of separation technologies.
Prerequisite: CHE 301.

Learning Outcomes

By the end of this course students will be able to:

- Be familiar with separation principles, design, separation mechanisms, processes, and applications
- Understand the underlying thermodynamic and mass transfer principles
- Understand the operation of simple and multistage batch distillation systems, and McCabe-Thiele method
- Discuss the differences between batch and continuous operation
- Explore use of various types of separation techniques
- Explore use of alternative separation techniques to the existing ones

Textbook and General References

Textbook

- Wankat, P. C., *Separation Processes Engineering: Includes Mass Transfer Analysis*, 4th edition, Prentice Hall, 2017

References

- Seader, J.D et al., *Separation Processes Principles*, 3rd edition, John Wiley & Sons, New Jersey, 2011

Office Hours:

TA: Jannice Lee (EIB264;jlee548@uic.edu) & Yuechen Gao (EIB241;ygao54@uic.edu)

TA office hours: Tuesday and Thursday, 1:00 – 3:00 pm

Professor office hours: Tuesday and Thursday, 12:20-1:20 pm

Course Website:

The course syllabus, homework assignments, and all handouts will be posted on the blackboard. Students will be responsible for keeping up to date with the information in this website

Grading Procedure:

Homework	10%
Quizzes	15%
Exam1	25%
Exam2	25%
Final Exam	25%

Homework:

Homework will be assigned every week along with the due date. You are encouraged to work through all the problems so that you can understand the materials and prepare themselves for the quizzes and exams. Final answers and compact solutions will be posted in the blackboard. But detailed written solutions will not be provided. You may consult anyone when working these problems. However, direct copying of another student's work will be considered an honor code violation. *<u>Without medical excuse, late homework will not be accepted</u>*.

Quizzes:
Quizzes will be occasionally given to encourage progressive mastery of the course content. There will be total five in-class quizzes. A student who is absent (regardless of the reason) will not be given a make-up quiz. All quizzes are closed-book and closed-notes.

Exams:

Exams and quizzes will cover all material up to and including the last lecture before the test, unless otherwise specified. Each test will emphasize the material since the last test. There will be no make-up exams. All exams are closed-book and closed-notes

Course Outline

- Introduction
  - Course outline and assessment
  - Introduction to separation processes
- Flash Distillation
- Binary distillation
- Multicomponent distillation
- Batch distillation
- Stage and packed column design
- Absorption and stripping
- Liquid-liquid extraction
- Membrane separations